Capstone Project – I: Review – 2 on

Energy Efficiency Optimization using IoT-Driven Power Consumption Analytics

Presented by the Batch No. : XXX

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Recap from review-1

- Presented the problem of inefficient energy consumption in households/industries.
- Showed literature survey on IoT + ML-based energy optimization.
- Demonstrated initial understanding of clustering & forecasting concepts.
- Received feedback to refine research gaps and provide a clearer execution roadmap.

Problem Statement

Traditional energy systems lack real-time monitoring, behaviour analysis, and consumption forecasting.

This makes it difficult to:

- Identify inefficient energy usage behaviour
- Predict future power demand
- Optimize appliance schedules
- Reduce electricity cost and wastage

Need: A unified IoT + ML system for real-time monitoring and predictive analytics.

Objectives

Phase 1 – Machine Learning Model Development

1. Build ML models to analyze historical consumption patterns (Clustering).
2. Develop an LSTM-based forecasting model for next-hour energy prediction.

Phase 2 – IoT System Integration

1. Integrate ESP32 + PZEM-004T sensors for real-time data acquisition.
2. Connect ML models to IoT pipeline for live predictions & recommendations.

Feedback from Guide/Reviewers and actions taken after Review–1

Feedback

1. Research gaps not clearly identified
2. No clear execution roadmap shown
3. Need clarity on ML techniques & architecture

Actions Taken

1. Research gaps refined (dataset limitations, lack of real-time systems, missing multivariate forecasting).
2. Detailed project workflow prepared (from dataset → ML → IoT → dashboard).
3. Implemented clustering model + started LSTM model for forecasting.

Requirement Analysis

- **Hardware Requirements**
  - ESP32 microcontroller
  - PZEM-004T energy sensor
  - CT coil / voltage sensor module
  - Wi-Fi router
  - Laptop/PC for ML model deployment

- **Software Requirements**
  - Python (Pandas, NumPy, Scikit-Learn, TensorFlow/Keras)
  - MQTT or REST API
  - Google Colab
  - Database: Firebase
  - Dashboard: Streamlit / React

Identification of tools/technology/algorithms

Tools

- Python, Jupyter Notebook
- Google Colab
- VS Code
- ESP32 Arduino IDE

Identification of tools/technology/algorithms

Technologies

- IoT Sensor Integration (ESP32 + PZEM-004T)
- MQTT / HTTP communication
- Machine Learning (Clustering, LSTM)

Identification of tools/technology/algorithms

Algorithms

- K-Means Clustering – detect behaviour patterns
- LSTM Deep Learning Model – demand forecasting

Design Strategy

Project Workflow

Phase 1 : Model Building

Data Source
(Public Dataset (UCI / IoT building data) , Features: Voltage, Current, Power, kWh)

↓

Data Preprocessing & Feature Eng.
(Missing value handling , Compute kWh (if missing) , Train–Test Split (80/20))

↓

Model Development
A. Clustering / Anomaly Detection
(K-Means / Autoencoder , Identify inefficient behaviour)

B. Forecasting
(LSTM / Prophet , Predict future Power / kWh)

C. Optimization & Scheduling
(Rule-based / RL Agent,Recommend energy-efficient times)

↓

Evaluation & Visualization
(Accuracy,Plots: Actual vs Predicted Power,Efficiency Scores / Recommendations)

Phase 2 : Extedning to IOT

IoT Layer Sensors:
((SCT-013 + ZMPT101B + ESP32 Wifi Module),Measures Voltage, Current, Power, kWh, Sends data via Wi-Fi (MQTT / HTTP))

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Edge / Gateway Layer:
(Receives sensor data, does minimal preprocessing , Sends formatted JSON to ML backend
{ timestamp, voltage, current, power, kwh })

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ML Processing Backend:
(Loads trained LSTM model (from Phase 1)
Performs: (Forecast(Power/kWh) , Inefficiency detection, Recommendation generation) Stores results in DB (MongoDB))

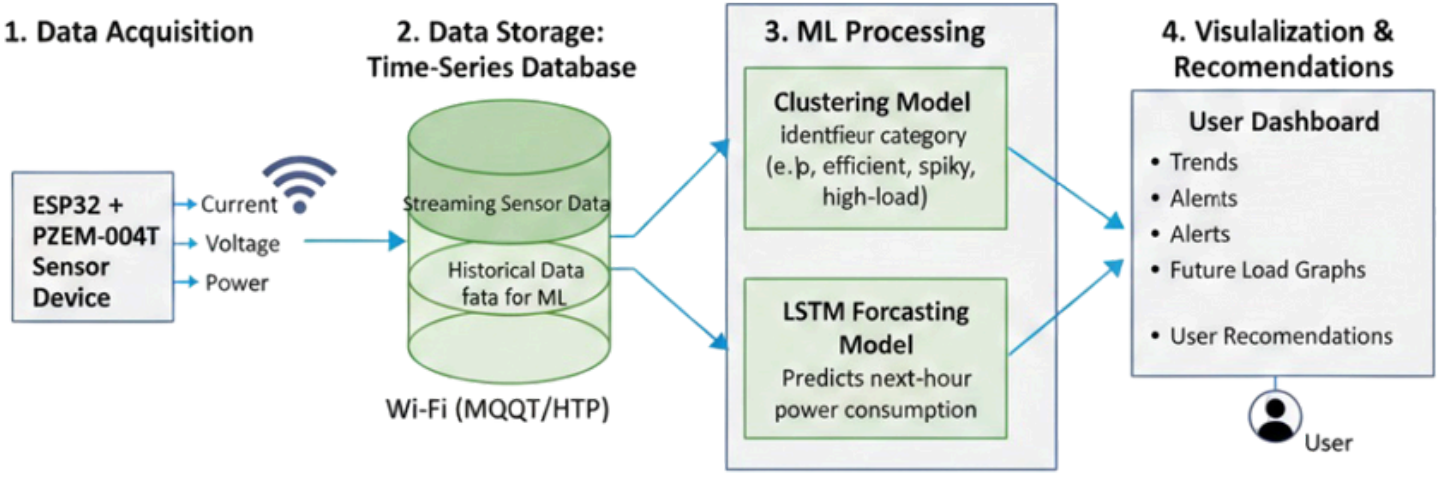
↓

Visualization Layer:
(Web Dashboard (React / Streamlit)
Displays: Real-time readings Forecasted load curve Alerts & recommendations Efficiency index)

Design Strategy

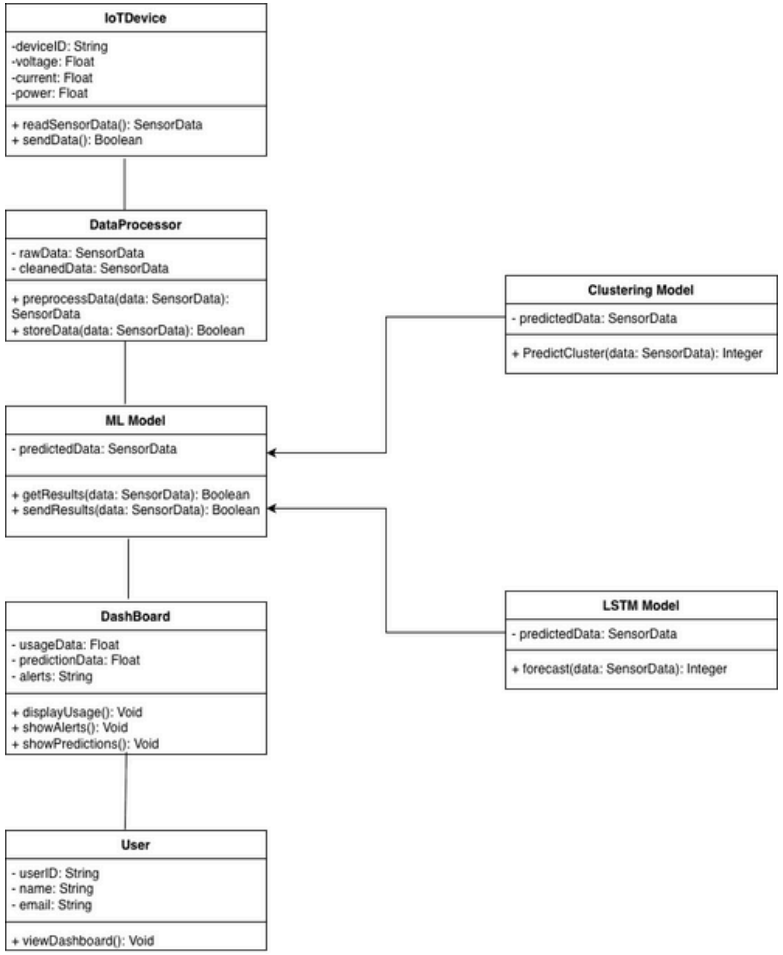
System Design

- Components:
  - IoT Layer: ESP32, PZEM sensor
  - Data Layer: local storage / Firebase
  - ML Layer:
  - Clustering model for behaviour segmentation
  - LSTM model for forecasting
  - Application Layer: Web/Mobile dashboard
  - Recommendation Engine: Suggests optimal usage times



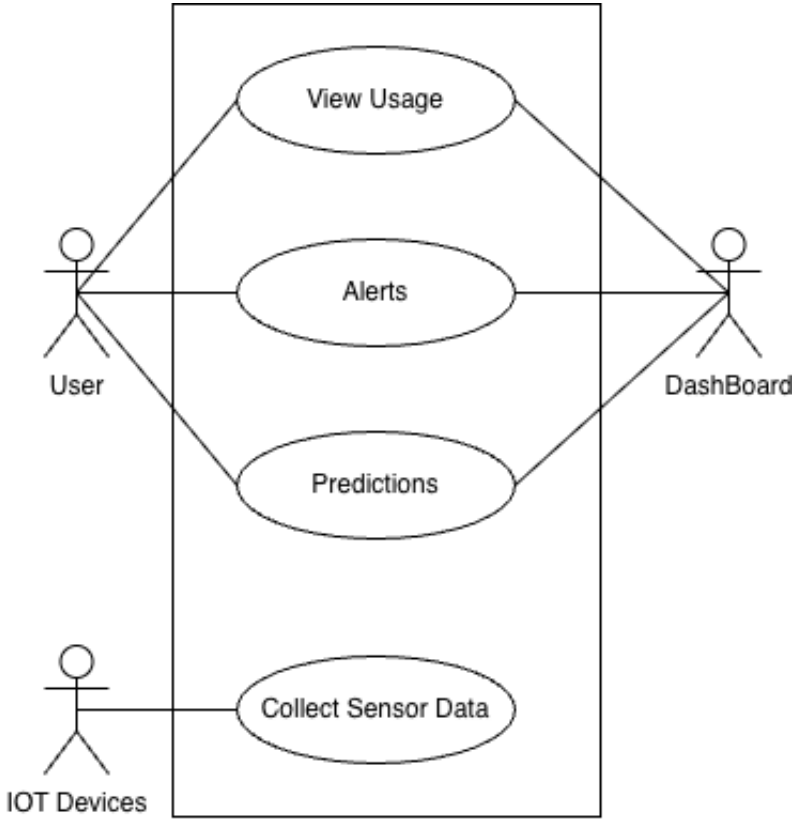
Design Strategy

Class Diagram



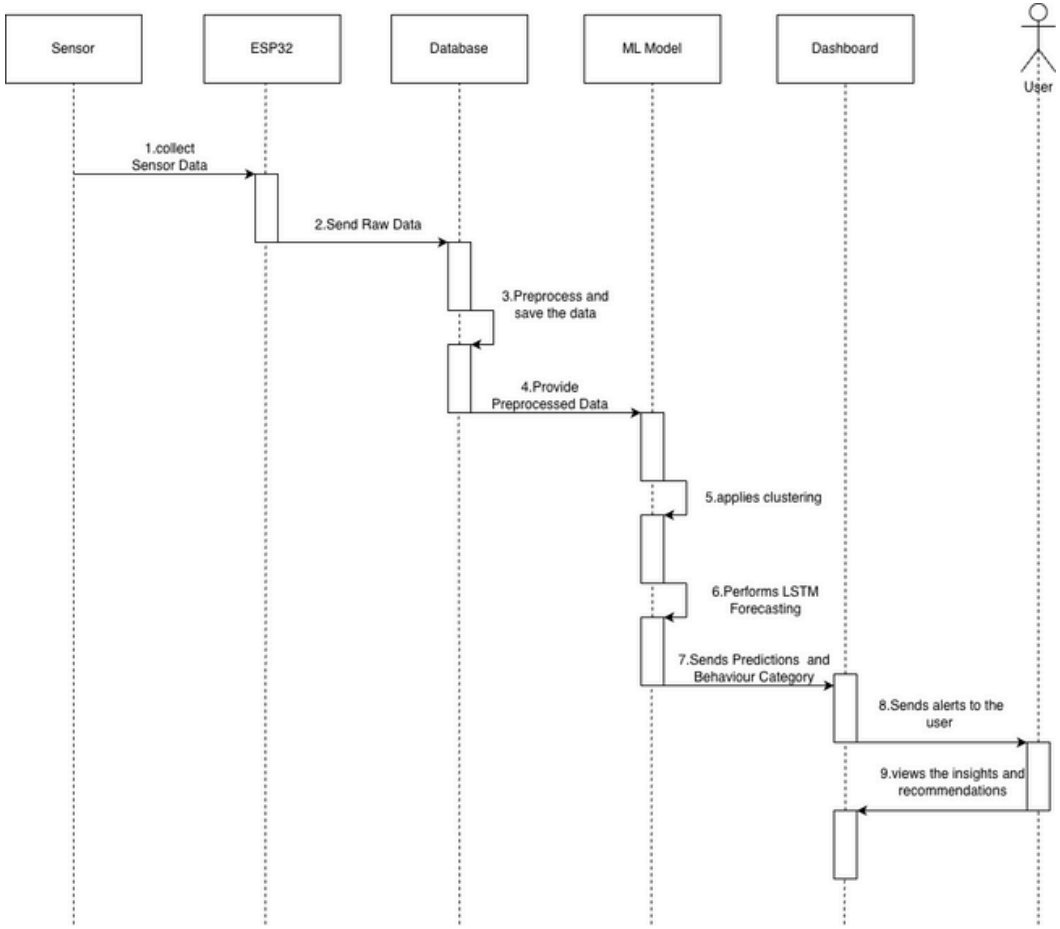
Design Strategy

UseCase Diagram



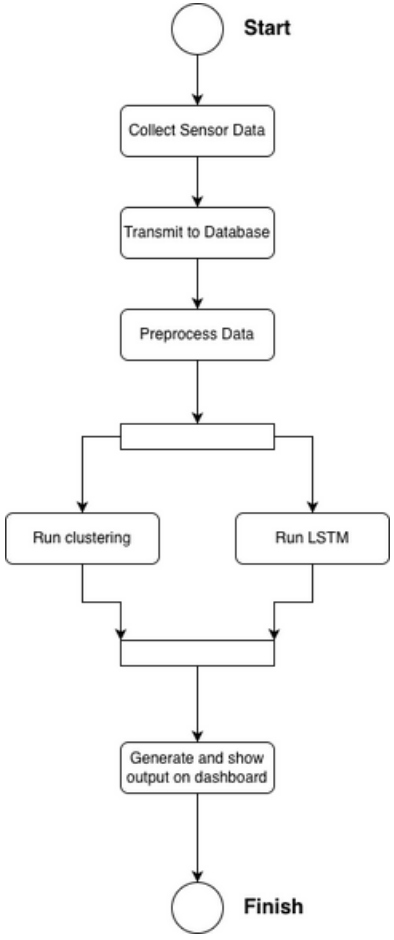
Design Strategy

Sequence Diagram



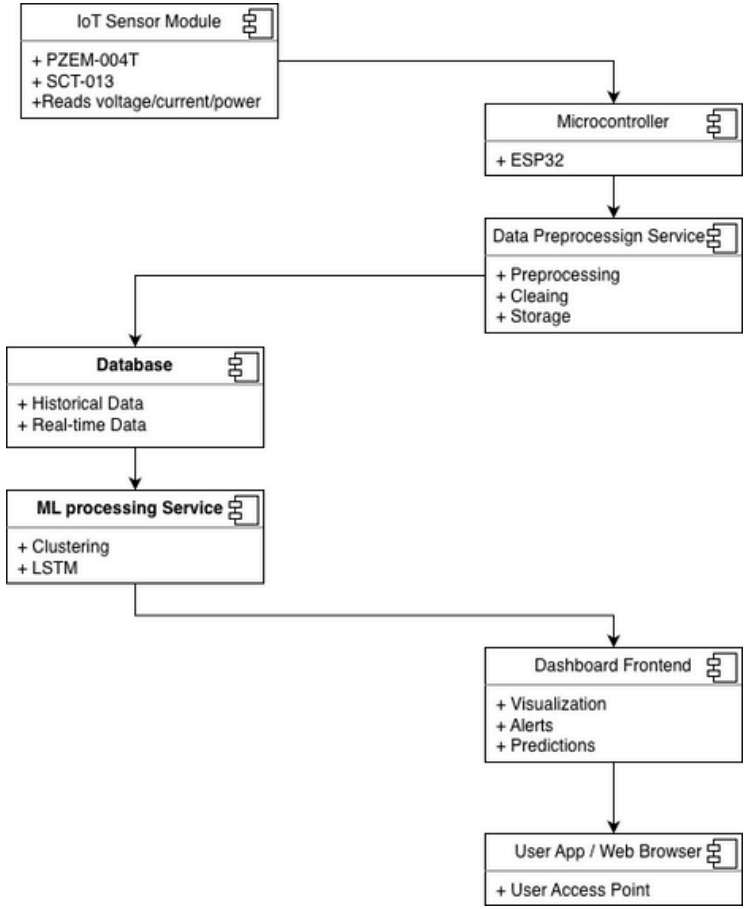
Design Strategy

Activity Diagram



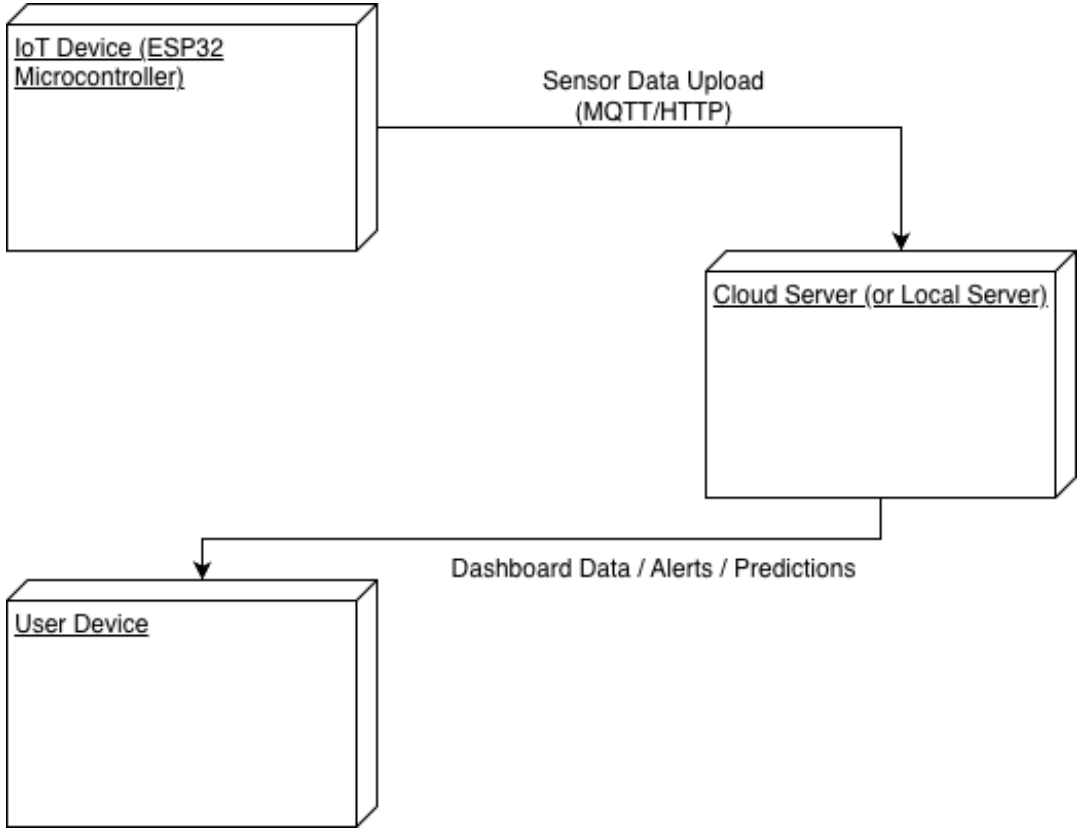
Design Strategy

Component Diagram



Design Strategy

Deployment Diagram



Conclusion

- Completed literature survey & identified strong research gaps
- Clustering model implemented successfully
- LSTM future consumption model partially completed
- Clear plan for IoT integration and dashboard development

References

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[3] PZEM-004T & SCT-013 Sensor Datasheets.

[4] Energy Optimization Research Papers (IoT, ML, Smart Grids)

[5] TensorFlow and Scikit-Learn Documentation

[6] Clustering Model Colab Link(https://colab.research.google.com/drive/1-U-GMgeKRNzcrYV4kobnuwHlzqKPd7IP)

[7] LSTM Model Colab Link(https://colab.research.google.com/drive/1-U-GMgeKRNzcrYV4kobnuwHlzqKPd7IP)